# *Hypoxia area vs. volume: Assessing hypoxia severity in the Gulf of Mexico*

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## Abstract

The northern Gulf of Mexico has dealt with the seasonal 'Dead Zone' for decades, an area of hypoxia (dissolved oxygen below 2 mg/L) occurring every summer that is driven by nutrient rich runoff from the Mississippi River (Rabalais et al., 2002). The occurrence of this hypoxic zone leads to alterations in organism behavior such as avoidance by nekton and reduced feeding (Rabalais et al., 2002). Organisms without the ability to avoid hypoxia, such as benthic invertebrates, suffer die offs and community shifts towards species that are generally opportunistic and have hypoxia tolerance (Rabalais & Turner, 2019). These benthic community shifts have effects of fisheries via food web shifts (Rabalais et al., 2002) and changes to biogeochemical cycling from decreased bioturbation (Rabalais & Turner, 2019). Attempts to understand and manage this annual disturbance led to the formation of the Mississippi River/Gulf of Mexico Hypoxia Task Force (2001) and the establishment of a 5-year average goal for hypoxia area. Climate change impacts are expected to lead to stronger stratification in the water column (Rabalais et al., 2010). These shifts are likely to impact the volume of the hypoxia zone as well as the area. While some studies have looked at volume of hypoxia (Matli & Obenour, 2024; Obenour et al., 2013), these studies used geostatistical model simulations and included no years beyond 2018. This study aims to use interpolation-based methods to estimate hypoxia volume and investigate and trends of volume over time and between the area and volume of hypoxia.

## Introduction

While hypoxia (defined as dissolved oxygen below 2 mg/L) is not an uncommon occurrence in the world's oceans and exists in a variety of areas, coastal hypoxia appears to be increasing and driven by human activities (Rabalais et al., 2002). The seasonal 'Dead Zone' in the northern Gulf of Mexico is one of largest human-caused areas of coastal hypoxia (Rabalais & Turner, 2019). Excess nutrients builds in the Mississippi River via runoff and is discharged into the northern Gulf of Mexico, leading to phytoplankton blooms that sink upon death and lead to the oxygen depletion via the decomposition process (Rabalais et al., 2002). Oxygen is unable to become reintroduced to the bottom waters due to stratification of the water column, driven by temperature and salinity.

Drops in oxygen levels result in shifts to the community composition in the area. Pelagic, and some demersal fish and mobile benthic organisms, suffer displacement and loss of suitable habitat (Rabalais et al., 2002). The benthic invertebrate community sees a general shift towards opportunistic and low-oxygen tolerant species and a reduction in biomass and abundance that remains after hypoxia dissipates, impacting food resources for demersal fish as the return leading to lower secondary production (Baustian & Rabalais, 2009; Rabalais & Turner, 2019).

Along with food web impacts, these changes in benthic invertebrate communities lead to less bioturbation and cause changes to the biogeochemical cycling via reduction to sediment mixing (Rabalais et al., 2010).

Since 1985, an annual research cruise began to map the extent of the hypoxia area in the Gulf of Mexico (Rabalais et al., 2002) following predetermined transects from the mouth of the Mississippi River to the Texas coast (Figure 1). A CTD instrument is lower at every station to measure water parameters of the water column and is used for the area estimations (Woods, n.d.). Hypoxia area has been predicted within acceptable error in years without storm disruption, with notable exceptions of 2022 (56% of predicted size) and 2023 (82% of predicted size) (Turner et al., 2024). Investigation into what variables the models needed to refine led researchers to the conclusion that beyond a certain threshold there is a 'temperature tipping point' where warming waters lead to decreased hypoxia area (Turner et al., 2024). This was found to be primarily related to the relation of cell size and growth rate to temperature and the slow sinking rate of smaller cells, leading to less organic material making its way to the bottom waters (Turner et al., 2024).

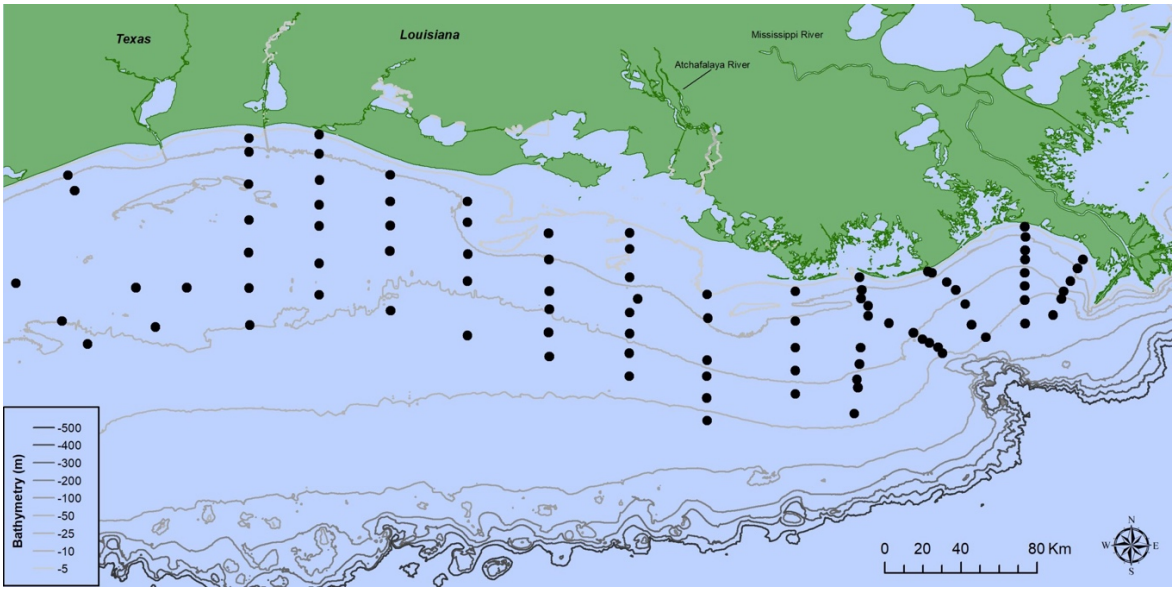



*Figure 1. Map of stations visited during the 2021 Annual Hypoxia cruise. Image from gulfhypoxia.net*

A few studies have conducted volume estimates using geostatistical model simulations, but the latest year included in these is 2018 (Matli & Obenour, 2024; Obenour et al., 2013; Scavia et al., 2019). This study aims to investigate more recent years of hypoxia volume estimates to identify any trends between years with similar area estimates. With the identification of temperature being able to go beyond a threshold and cause area declines, it is important to understand if a similar trend holds true for volume. Volume based estimations will provide better understanding and predictions for ecological impacts of hypoxia in the northern Gulf of Mexico. This study also aims to investigate any potential trends in the extent of hypoxia to the east of the Mississippi River Delta, which was included in the annual mapping in 2024 and 2025.

# Methods and Timeline

## *Weeks 1 - 2*

CTD profile data from the annual Gulf of Mexico hypoxia cruise for target years will be compiled. Priority years will include the two years that included the east side of the river delta and years with similar area estimates. This will include ensuring data is correctly formatted and can be uploaded into Python. Area estimates for selected years will be compiled into a single file. A master list of station coordinates will be made and the coordinate reference system used for it will be verified.

A workflow will be developed for performing map interpolation and hypoxia volume calculations. The reported area maps are created using ArcGIS Pro Topo to Raster tool. To create the volume estimates in as similar a way as possible to the area, the ArcPy package will be used. ArcPy provides access to the ArcGIS Pro tools within the Python environment. Workflow development will focus on using ArcPy for mapping and interpolation of the data.

## *Weeks 3 - 4*

Using the developed mapping and interpolation workflow, the volume estimations for the target years will be calculated. These estimates will be compiled and used to compare to the reported area from the same year. Maps of volume estimates will be generated using ArcPy. Linear regression analysis will be done on hypoxia area and volume to investigate any correlations between the two variables and any trends of volume over time. The SciPy package, specifically the scipy.stats() module, will be used to perform the linear regression analysis.

## *Weeks 5 - 6*

The final two weeks will focus on the interpretation of results. The primary focus will be writing the final report and creating the final presentation.

If adequate time remains after the initial round of data processing and analysis in weeks 3 – 4, more cruise years will be added to the dataset and wind data from buoy data off the Louisiana coast will be incorporated. Linear regression analysis will be used to investigate any correlations between changes to wind velocities and hypoxia extent.

# Broader Impacts

Hypoxia in the northern Gulf of Mexico produces a recurring stressor on the ecosystem and impacts fisheries via food web disruptions and habitat squeeze (Rabalais et al., 2002). However, despite decades of mapping, there is still much to understand about the drivers behind the extent of hypoxia formed from the nutrient-rich river discharge every year (Turner et al., 2024). Understanding of hypoxia volume provides insight into the potential ecological impacts that observing area alone cannot.

Figure 2 shows CTD profiles of two stations from two different years. Station E-F5 is from the east of the river sites and shows a drastic difference in percent of the water column below the hypoxia threshold between 2024 and 2025. With 2024 not just being hypoxic, but only the top roughly 8 m of water being above the threshold. This limited vertical space has major implications for the movement of pelagic fish and is not something caught by area mapping.

Station A6 profiles are from 2015 and 2021, two years that had similar area estimates. The CTD profiles show not only a roughly 17 m difference in hypoxia, but for both years the bottom waters remained above the hypoxia threshold of 2 mg/L. This mid-column hypoxia is not something that is accounted for by the annually produced area maps. It does, however, have ecological consequences. Masses in the middle of the water column like what is shown in Figure 2 can cause disruptions in vertical migrations of plankton, leading to cascading trophic effects. Understanding the full impacts of hypoxia on ecology in the Gulf of Mexico requires considering, not just its areal variability, but volume as well.

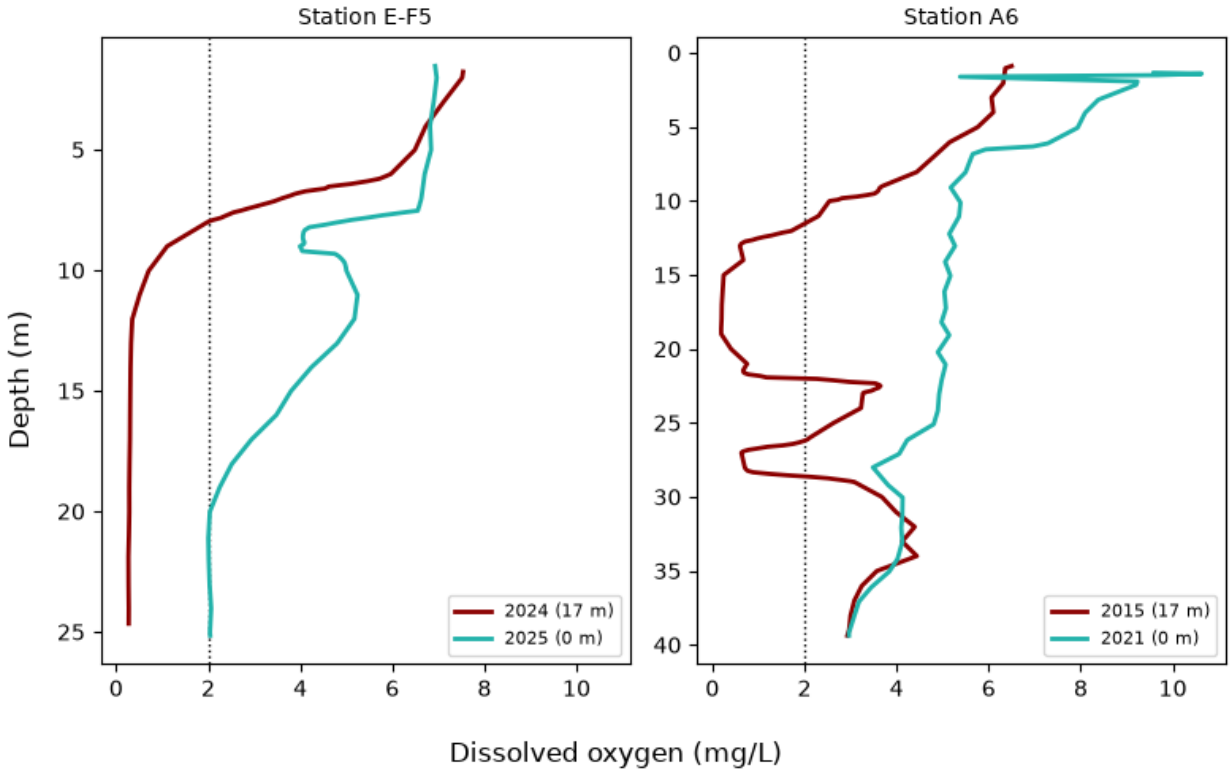



*Figure 2. CTD profiles of two stations in two different years. Depth (m) on the Y-axis, dissolved oxygen (mg/L) on the X-axis. Hypoxia threshold marked by dashed grey line. Year with higher percent of water column hypoxic in red, lower in blue. Data accessed from NOAA database.*

# References


Baustian, M. M., & Rabalais, N. N. (2009). Seasonal Composition of Benthic Macroinfauna Exposed to Hypoxia in the Northern Gulf of Mexico. *Estuaries and Coasts*, *32*(5), 975–983. https://doi.org/10.1007/s12237-009-9187-3

Matli, V. R. R., & Obenour, D. (2024). Trends and drivers of hypoxic thickness and volume in the northern Gulf of Mexico: 1985–2018. *PLOS ONE*, *19*(12), e0302759. https://doi.org/10.1371/journal.pone.0302759

Mississippi River Nutrient/Gulf of Mexico Hypoxia Task Force. (2001). *Action plan for reducing, mitigating, and controlling hypoxia in the northern Gulf of Mexico*. Washington, D.C.: Office of Wetlands, Oceans, and Watersheds, U.S. Environmental Protection Agency.

Obenour, D. R., Scavia, D., Rabalais, N. N., Turner, R. E., & Michalak, A. M. (2013). Retrospective Analysis of Midsummer Hypoxic Area and Volume in the Northern Gulf of Mexico, 1985–2011. *Environmental Science & Technology*, *47*(17), 9808–9815. https://doi.org/10.1021/es400983g

Rabalais, N. N., Díaz, R. J., Levin, L. A., Turner, R. E., Gilbert, D., & Zhang, J. (2010). Dynamics and distribution of natural and human-caused hypoxia. *Biogeosciences*, *7*(2), 585–619. https://doi.org/10.5194/bg-7-585-2010

Rabalais, N. N., & Turner, R. E. (2019). Gulf of Mexico Hypoxia: Past, Present, and Future. *Limnology and Oceanography Bulletin*, *28*(4), 117–124. https://doi.org/10.1002/lob.10351

Rabalais, N. N., Turner, R. E., & Wiseman, W. J. (2002). Gulf of Mexico Hypoxia, A.K.A. "The Dead Zone." *Annual Review of Ecology and Systematics*, *33*(1), 235–263. https://doi.org/10.1146/annurev.ecolsys.33.010802.150513

Scavia, D., Justić, D., Obenour, D. R., Craig, J. K., & Wang, L. (2019). Hypoxic volume is more responsive than hypoxic area to nutrient load reductions in the northern Gulf of Mexico—And it matters to fish and fisheries. *Environmental Research Letters*, *14*(2), 024012. https://doi.org/10.1088/1748-9326/aaf938

Turner, R. E., Rabalais, N. N., & Glaspie, C. N. (2024). A temperature tipping point in hypoxic zone size. *Limnology and Oceanography*, *69*(12), 2954–2962. https://doi.org/10.1002/lno.12722

Woods, G. (n.d.). Shelfwide Cruises. *Gulf Hypoxia*. Retrieved October 22, 2026, from https://gulfhypoxia.net/research/shelfwide-cruises/